

SHRISH KUMAR SINGHAL

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SUMMARY

Quantitative analyst who builds models, validates them against alternatives, and runs them in production. Prices instruments more than one way and treats the disagreement as the finding: option valuation implemented three independent ways as mutual cross checks, and a Treasury curve bootstrap producing the discount and forward rates a swap is valued off. Equally hands-on with modern machine learning, including retrieval augmented generation, agentic systems and parameter efficient fine tuning, applied with the same benchmark-and-document discipline. MS in Computer Science from NYU, B.Tech from IIT Guwahati, FRM Part I passed in the top quartile of all four domains.

EDUCATION AND CERTIFICATION

Financial Risk Manager (FRM), Part I

Passed August 2026

Global Association of Risk Professionals | Digital Badge

- Top quartile (76th to 100th percentile) in **all four** exam domains: Foundations of Risk Management; **Quantitative Analysis**; **Financial Markets and Products**; **Valuation and Risk Models**.

New York University, Tandon School of Engineering

Sept 2023 - Jan 2025

Master of Science in Computer Science | GPA: 3.83 / 4.0

New York, NY

- Coursework: Quantitative Finance, Machine Learning, Deep Learning, Artificial Intelligence, Big Data, Data Engineering

Indian Institute of Technology (IIT) Guwahati

July 2019 - May 2023

B.Tech. Electronics and Communication Engineering; Minor in Computer Science

Guwahati, India

- Admitted via JEE Advanced at rank **1,395 of 1.5 million** candidates, top 0.1 percent nationally.

EXPERIENCE

Data Scientist, then AI Engineer

June 2025 - August 2026

Boson Motors

San Jose, CA

- Owned a **daily production process end to end** over **50,000+** records per hour, from data collection through generation to quality control, with completeness and staleness tests on each feed, distributional checks on output, and cross-source reconciliation that removed **100%** of a recurring manual verification step.
- **Investigated variance and anomalies to root cause** rather than correcting the symptom, tracing suspect figures back through the transformation chain, escalating when issues were material, and adding controls that prevented the same breaks from recurring.
- **Developed, tested, documented, and monitored models in live use**, with automated retraining, drift detection between expected and realized behavior, and documented validation checks on every rebuild, sustaining above **95%** accuracy in production.
- Selected estimation techniques based on evidence: paired **classical statistical methods** (Isolation Forest, distributional tests) with **machine learning** (LSTM autoencoders) and benchmarked them against each other. Delivered forecasts that predicted failures two weeks ahead and cut unplanned downtime by **34%**, alongside a reinforcement learning framework (Soft Actor-Critic) calibrating 10+ parameters for a **25%** improvement.
- Built **LLM systems**: developed an agent with authenticated, permission-scoped access to internal platform APIs, and deployed a **retrieval-augmented generation** pipeline over technical documentation, with outputs validated against source data rather than trusted implicitly.
- **Communicated to non-technical stakeholders** daily, translating methods, assumptions, and failure modes into actionable insights for engineering, operations, and management, while building the dashboards and written summaries they relied on.

Quantitative Researcher

June 2024 - Oct 2024

New York University | Supervisor: Prof. Carlos De Oliveira

New York, NY

- Designed and implemented a **statistical arbitrage strategy** across the NASDAQ, S&P 500 and Dow Jones over **24 years** of daily data, writing the full research stack in Python: ingestion and validation, feature construction, a custom back testing engine, and daily P&L, exposure and risk reporting. **12% CAGR in backtest, net of 0.5% assumed transaction costs**.
- Researched **US Treasury rates as a macro driver**, forecasting the 10 year yield over a 100 day forward horizon and integrating it through SQL and pandas to condition positioning on the rate environment, then **isolated its marginal contribution** at roughly 2 percentage points of annual return rather than assuming it helped.
- **Implemented an alternative benchmark and reported the comparison honestly**, running a classical statistical specification (SARIMA) against a machine learning one (LSTM) on identical data and stating which won.
- Evaluated across **18 consecutive quarters** chosen to include the 2020 drawdown and the 2022 rate hiking cycle, reported the worst quarter at **-9.9%** alongside the headline figure, and built a risk overlay that adjusts exposure when realised risk breaches a calibrated tolerance.

- Conducted an **independent benchmark review**: assessed four published methods, implemented an alternative, and evaluated all five on a common dataset, reporting both where the alternative won (best Jaccard 0.8729, best recall 0.9662 of the set) and **where it lost** (Dice 0.9321 against 0.9396 for a fully trained convolutional model, at a fraction of its compute).

PROJECTS

- **US Treasury Curve, Carry and Roll-Down** (Python, pandas, NumPy, SciPy) | **2026**
 - Collects the H.15 constant-maturity curve and SOFR from FRED daily, then **bootstraps a zero curve** from the par yields one coupon date at a time, producing the discount factors, zero rates and forward rates that a vanilla interest rate swap is valued off.
 - Prices repo-financed cash Treasuries through **cash-and-carry**, decomposing carry and roll-down exactly rather than by first-order approximation. Extends to DV01-neutral spread and fly economics.
 - **Sixteen self-checks** on the maths: every input par yield reprices its own bond to 100 within 1e-9, cash-and-carry at the curve-implied repo reproduces the curve's own forwards, and trade signs are pinned to explicitly constructed DV01-neutral P&Ls after an earlier version had the convention backwards.
- **Option Pricing and Cross Validation** (Python, NumPy, pandas) | **2024**
 - Values European and American options three independent ways, **Black-Scholes closed form, Cox-Ross-Rubinstein binomial lattice and Monte Carlo simulation**, using each method as an accuracy check on the others, with automated P&L, scenario and risk exposure summaries.
- **Model Comparison and Back Testing Framework** (Python, pandas, NumPy) | **2024**
 - Abstract interface where a candidate model is specified once and the engine handles ingestion, screening for stale instruments, exposure sizing against a volatility budget and daily accounting, so competing specifications are compared on identical data. Known defects documented in the repository, including survivorship bias and one sizing input computed but not yet applied.
- **Parameter Efficient Fine Tuning and Evaluation** (PyTorch, LoRA, Unsloth) | **2024**
 - Supervised fine tuning of an 8B parameter model with low rank adapters and 4 bit quantisation, **evaluated at 82.5%** on a reasoning benchmark, with training time cut **20%** on a single consumer GPU through gradient checkpointing and memory profiling.

QUANTITATIVE AND TECHNICAL CAPABILITY

- **Valuation and derivatives**: option pricing theory implemented from first principles across closed form, binomial lattice and Monte Carlo; forward pricing and cash-and-carry; duration, convexity and DV01; P&L and exposure attribution.
- **Rates and curve construction**: par-to-zero bootstrapping, discount factors, spot and forward rates, forward par yields, carry and roll-down decomposition, repo financing and implied repo, macroeconomic regime conditioning on the 10 year Treasury.
- **Statistics and econometrics**: probability theory, real analysis, stochastic and scientific computing, detection and estimation theory; hypothesis testing, regression, time series (ARIMA, SARIMA, LSTM), Monte Carlo simulation, scenario and sensitivity analysis.
- **Machine learning and AI**: regression, classification, clustering, anomaly detection (autoencoders, Isolation Forest), deep learning (LSTM, CNN, attention), reinforcement learning (Soft Actor Critic), transformers, LoRA fine tuning and quantisation, retrieval augmented generation, agentic systems with scoped tool access, model evaluation and drift monitoring.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, PyTorch, TensorFlow, Polars), **SQL**, R, MATLAB, LaTeX; modular and reusable design, version control and code review with Git and GitHub, Docker, AWS, Linux

Data: ETL design, data quality validation and reconciliation, exception detection, large and unstructured datasets, distributed processing (Spark, Spark Streaming, PySpark, Polars, 100GB+), relational and NoSQL stores, vector search

Reporting: Tableau, Plotly, Streamlit, Grafana, dashboard building for non-technical readers, Microsoft Excel (lookups, pivot tables), PowerPoint, written technical documentation

PUBLICATIONS AND AWARDS

- S. K. Singhal, C. De Oliveira. “**An LSTM Case Study for Trading American Liquid Indices.**” *Working paper, New York University*, 2024.
- S. K. Singhal et al. “**Segmentation of Bone Tissue from CT Images.**” *Springer Lecture Notes in Networks and Systems*, vol. 586, pp. 233 to 242, 2023. And “**Security Analysis of Aadhaar Authentication Process and Way Forward**”, *IEEE ICAC3N* 2021, sole author.
- **Academic:** JEE Advanced rank 1,395 of 1.5M candidates (top 0.1%); KVPY Science Fellow 2018 and 2019 (top 0.5% of 50,000); NTSE Scholar; Mitacs Globalink Research Fellow, Canada 2022. Top 10 finalist of 2,700+ teams, National Aadhaar Hackathon 2021.